User's Guide for the amsmath Package (Version 2.1)

American Mathematical Society, LaTeX Project

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Introduction

The amsmath package is a LaTeX package that provides miscellaneous enhancements for improving the information structure and printed output of documents that contain mathematical formulas. Readers unfamiliar with LaTeX should refer to [?]. If you have an up-to-date version of LaTeX, the amsmath package is normally provided along with it. Upgrading when a newer version of the amsmath package is released can be done via `http://mirror.ctan.org/macros/latex/required/amsmath.zip`.

This documentation describes the features of the amsmath package and discusses how they are intended to be used. It also covers some ancillary packages:

```
amsbsy   amsopn    amsxtra
amscd    amstext
```

These all have something to do with the contents of math formulas. For information on extra math symbols and math fonts, see [?] and `https://www.ams.org/tex/amsfonts.html`. For documentation of the `amsthm` package or AMS document classes (`amsart`, `amsbook`, etc.) see [?] or [?] and `https://www.ams.org/tex/author-info.html`.

If you are a long-time LaTeX user and have lots of mathematics in what you write, then you may recognize solutions for some familiar problems in this list of amsmath features:

- A convenient way to define new 'operator name' commands analogous to `\sin` and `\lim`, including proper side spacing and automatic selection of the correct font style and size (even when used in sub- or superscripts).
- Multiple substitutes for the `eqnarray` environment to make various kinds of equation arrangements easier to write.
- Equation numbers automatically adjust up or down to avoid overprinting on the equation contents (unlike `eqnarray`).
- Spacing around equals signs matches the normal spacing in the `equation` environment (unlike `eqnarray`).
- A way to produce multiline subscripts as are often used with summation or product symbols.
- An easy way to substitute a variant equation number for a given equation instead of the automatically supplied number.
- An easy way to produce subordinate equation numbers of the form (1.3a) (1.3b) (1.3c) for selected groups of equations.

The amsmath package is distributed together with some small auxiliary packages:

amsmath Primary package, provides various features for displayed equations and other mathematical constructs.

`amstext` Provides a `\text` command for typesetting a fragment of text inside a display.

`amsopn` Provides `\DeclareMathOperator` for defining new 'operator names' like `\sin` and `\lim`.

`amsbsy` For backward compatibility this package continues to exist but use of the newer `bm` package that comes with LaTeX is recommended instead.

`amscd` Provides a `CD` environment for simple commutative diagrams (no support for diagonal arrows).

`amsxtra` Provides certain odds and ends such as `\fracwithdelims` and `\accentedsymbol`, for compatibility with documents created using version 1.1.

The amsmath package incorporates `amstext`, `amsopn`, and `amsbsy`. The features of `amscd` and `amsxtra`, however, are available only by invoking those packages separately.

The independent `mathtools` package [**?**] provides some enhancements to amsmath; `mathtools` loads amsmath automatically, so there is no need to separately load amsmath if `mathtools` is used. Some `mathtools` facilities will be noted below as appropriate.

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Options for the amsmath package

The amsmath package has the following options:

`centertags` (default) For an equation containing a `split` environment, place equation numbers vertically centered on the total height of the equation.

`tbtags` 'Top-or-bottom tags': For an equation containing a `split` environment, place equation numbers level with the last (resp. first) line, if numbers are on the right (resp. left).

`sumlimits` (default) Place the subscripts and superscripts of summation symbols above and below, in displayed equations. This option also affects other symbols of the same type—$\prod$, $\coprod$, $\bigotimes$, $\bigoplus$, and so forth—but excluding integrals (see below).

`nosumlimits` Always place the subscripts and superscripts of summation-type symbols to the side, even in displayed equations.

`intlimits` Like `sumlimits`, but for integral symbols.

`nointlimits` (default) Opposite of `intlimits`.

`namelimits` (default) Like `sumlimits`, but for certain 'operator names' such as det, inf, lim, max, min, that traditionally have subscripts placed underneath when they occur in a displayed equation.

`nonamelimits` Opposite of `namelimits`.

To use one of these package options, put the option name in the optional argument of the `\usepackage` command—e.g., `\usepackage[intlimits]{amsmath}`. For AMS document classes and any other classes that preload amsmath desired options must be specified with the `\documentclass`—e.g., `\documentclass[intlimits,tbtags,reqno]{amsart}`.

The amsmath package also recognizes the following options which are normally selected (implicitly or explicitly) through the `\documentclass` command, and thus need not be repeated in the option list of the `\usepackage{amsmath}` statement.

`leqno` Place equation numbers on the left.

`reqno` Place equation numbers on the right.

`fleqn` Position equations at a fixed indent from the left margin rather than centered in the text column.

Three options have been added to control the space to the left of `aligned` and `gathered` environments. Prior to the 2017 release a thin space was added to the left but not the right of these constructs. This appears to have been an accidental feature of the implementation and was typically corrected by prefixing the environments with `\!`.

The new default behavior is aimed to ensure that the environments do not have a thin space added in most cases, and that existing documents using `\!\begin{aligned}` continue to work as before.

`alignedleftspaceyes` Always add `\,` to the left of `aligned` and `gathered`.

`alignedleftspaceno` Never add `\,` to the left of `aligned` and `gathered`.

`alignedleftspaceyesifneg` Only add `\,` if the environment is prefixed by negative space. (New default behavior.)